

Notes: Finkelstein (QJE, 2009)

E-ZTax: Tax Salience and Tax Rates

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1 Big picture

- **Question.** Does reducing the salience of a tax instrument lead governments to set higher tax rates? The paper studies whether electronic toll collection (ETC), by making tolls less visible to drivers, changes toll rates.
- **Core setting.** The empirical setting is U.S. toll facilities. ETC systems such as E-ZPass allow drivers to pay automatically rather than handing over cash. This plausibly reduces the attention paid to the toll amount.
- **Key shock.** Different toll facilities adopted ETC at different times from 1987 through 2005, and some never adopted it. This staggered adoption creates within-facility variation in toll salience.
- **Main result.** Once ETC diffuses to its steady-state penetration rate, tolls are estimated to be about 20–40 percent higher than they would have been under manual toll collection.
- **Mechanism 1: behavioral elasticity.** Under ETC, driving becomes less elastic with respect to the toll. If drivers are less responsive to toll changes, the revenue-maximizing or Pigouvian toll rises.
- **Mechanism 2: political costs.** Under manual collection, toll increases are lower in election years. Under ETC, this election-year sensitivity largely disappears, consistent with lower political costs of raising less visible tolls.
- **Economic takeaway.** Tax design affects equilibrium policy, not only individual behavior. A less salient payment technology can change the political economy of tax setting.

2 Incremental contribution of the paper

- **From individual misperception to equilibrium tax policy.** Prior work shows that consumers may misperceive taxes when taxes are less salient. This paper asks whether governments respond by setting higher tax rates.
- **Clean institutional source of salience variation.** ETC adoption changes how the same toll is paid without changing the taxed activity itself. The staggered rollout across facilities creates a credible panel setting.
- **New historical data.** The author assembles a new facility-level panel of toll rates, ETC adoption dates, ETC penetration, traffic, revenue, and institutional features for U.S. toll facilities.
- **Two-channel evidence.** The paper does not only estimate the reduced-form toll increase. It also provides evidence for two mechanisms: reduced demand elasticity and reduced election-year political discipline.
- **Political economy of tax instruments.** The paper gives empirical content to the classic fiscal-illusion argument: a less visible tax system can permit higher taxes because taxpayers/voters notice less.
- **Broader policy relevance.** The results are relevant to debates about withholding, value-added taxes, automatic payment systems, and other tax instruments whose incidence or payment may be less visible.

3 Conceptual framework

3.1 Perceived versus actual taxes

Let τ denote the actual tax and $\tilde{\tau}$ the perceived tax. Tax salience is indexed by θ , where a larger θ means lower salience. A fully salient tax system satisfies $\tilde{\tau} = \tau$.

The paper writes the perceived tax at decision margin $j \in \{c, v\}$ as

$$\tilde{\tau}_j(\theta) = \delta_{0j}(\theta) + \delta_{1j}(\theta)\tau,$$

where c refers to consumption decisions and v to voting or political decisions.

Interpretation. A less salient tax weakens the link between the actual tax and the tax that people use in decision-making. This can matter both when people decide whether to drive and when they decide whether to punish politicians for toll increases.

3.2 Consumer response

The individual chooses the taxed good x_1 based on the perceived tax at consumption. With quasi-linear utility and constant elasticity demand, the paper derives the empirical specification

$$\Delta \log x_1 = \beta_1 \Delta \log(p + \tau) + \beta_2 \theta + \beta_3 \theta \Delta \log(p + \tau) + \Delta \varepsilon.$$

The key coefficient is β_3 . If $\beta_3 > 0$ and the baseline elasticity is negative, lower salience makes demand less elastic in absolute value.

Interpretation. If ETC makes drivers less responsive to toll changes, a toll authority can raise tolls with less traffic loss.

3.3 Government response

The government chooses the tax to balance an economic objective and political costs:

$$\max_{\tau} \lambda \tau Q(p + \tilde{\tau}_C) - (1 - \lambda) f(E) C(\tilde{\tau}_V),$$

where E indicates election years.

The sign of the relationship between salience and tax rates is theoretically ambiguous because salience can affect both demand elasticity and political costs. The paper therefore estimates this sign empirically.

Political prediction. If reduced salience lowers political costs, election-year suppression of toll increases should be weaker after ETC adoption.

4 Data and measurement

4.1 Toll-facility panel

The target sample is publicly owned U.S. toll facilities charging tolls in 1985. The final dataset covers 123 facilities run by 49 operating authorities in 24 states or state-like entities. On average, the data include about 50 years of toll rates per facility.

Facilities include roads, bridges, and tunnels. The main toll variable is the nominal passenger-car toll for a full-length trip on a road or a round trip on a bridge/tunnel.

4.2 ETC adoption and penetration

ETC adoption dates range from 1987 to 2005, with a median around 1999. By 2005, 87 of the 123 facilities had adopted ETC. Once ETC is adopted, usage diffuses gradually. ETC penetration reaches a steady-state level of about 60 percent after approximately 14 years.

4.3 Toll measures

The paper distinguishes between the manual toll and the electronic toll. Some facilities offer ETC discounts. The main analysis uses the minimum toll, defined as the lower envelope of the manual and electronic tolls. Robustness checks use the manual toll and a sample of facilities that never offered ETC discounts.

4.4 Traffic and revenue data

Traffic data are available for 76 facilities and revenue data for 45 facilities. These data allow the paper to estimate driving elasticity and check whether changes in traffic, revenue, and tolls precede ETC adoption.

4.5 Survey evidence on salience

The paper uses two surveys. The Massachusetts survey asks drivers at an antiques show about the toll they just paid on the Massachusetts Turnpike. The NY/NJ survey asks regular users of Port Authority crossings about the toll they paid. In both surveys, ETC users are substantially less aware of toll amounts than cash users.

5 Empirical design

5.1 Traffic elasticity specification

The demand-side regression is

$$\Delta \log(\text{traffic})_{it} = \gamma_t + \beta_1 \Delta \log(\text{minimum toll})_{it} + \beta_2 \Delta \log(\text{minimum toll})_{it} \times \text{NeverETC}_i \\ + \beta_3 \Delta \log(\text{minimum toll})_{it} \times \text{ETC Penetration}_{it} + \beta_4 \text{NeverETC}_i + \beta_5 \text{ETC Penetration}_{it} -$$

The key coefficient is β_3 . A positive coefficient means that as ETC penetration rises, the demand elasticity becomes less negative.

5.2 Toll-rate specification

The baseline toll-rate regression is

$$\Delta y_{it} = \gamma_t + \beta_1 \text{ETC Adopt}_{it} + \beta_2 \text{ETC}_{it} + \Delta \mu_{it}.$$

Here ETC_{it} is one in the year ETC begins, and ETC_{it} is one in the adoption year and all later years. The coefficient on ETC_{it} captures additional annual toll growth after ETC adoption.

5.3 ETC penetration specification

To test whether toll increases follow the diffusion of ETC, the paper estimates

$$\Delta \log(\text{minimum toll})_{it} = \gamma_t + \beta_1 ETC_{it} + \beta_2 \Delta ETC_{it} + \varepsilon_{it}.$$

This asks whether tolls rise more in years when ETC use rises more.

5.4 Political business cycle specification

The politics regression augments the toll-rate equation with election-year indicators:

$$y_{it} = \gamma_t + \beta_1 ETC_{it} + \beta_2 ETC_{it} + \beta_3 ElecYear_{st} + \beta_4 ElecYear_{st} \times ETC_{it} + \beta_5 ElecYear_{st} \times ETC_{it} + \varepsilon_{it}.$$

The key prediction is that $\beta_3 < 0$ under manual toll collection and that the interaction with ETC is positive.

5.5 Identification

The empirical strategy uses within-facility changes around ETC adoption and year fixed effects. Identification requires that ETC adoption timing not be driven by unobserved toll-growth shocks. The paper addresses this by checking robustness across regions, facility types, facility age, facility fixed effects, infrastructure controls, and pretrends in traffic, revenue, and tolls.

6 Tables and figures

6.1 Table I and Figure I: Salience survey and ETC adoption timing

Read:

- Table I shows that ETC users are much less aware of tolls than cash users. In the MA survey, 61.8 percent of ETC users initially report that they do not know the toll, compared with 2.1 percent of cash users.
- ETC users are also more likely to estimate the toll incorrectly and make larger errors when they misreport.
- Figure I shows that ETC adoption dates are staggered from the late 1980s through 2005.

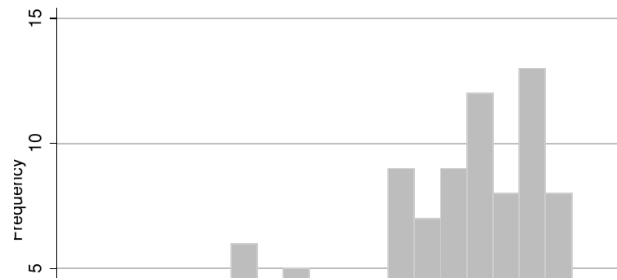
Economic message: Table I validates the maintained assumption that ETC reduces toll salience. Figure I shows the staggered adoption timing that underlies the panel identification strategy.

TABLE I
SURVEY EVIDENCE ON DRIVER AWARENESS OF TOLLS, BY PAYMENT METHOD

	MA survey				NYNJ survey		
	Difference between ETC and cash drivers				Difference between ETC and cash drivers (no covariates)		
	ETC drivers (1)	Cash drivers (2)	No covariates (3)	Covariate adjusted (4)	ETC drivers (5)	Cash drivers (6)	(7)
Fraction who report "don't know"	0.618 (0.490)	0.021 (0.142)	0.597*** (0.060)	0.579*** (0.060)	0.381 (0.486)	0.200 (0.400)	0.18*** (0.05)
Fraction who incorrectly estimate toll	0.851 (0.359)	0.308 (0.463)	0.543*** (0.058)	0.512*** (0.067)	0.826 (0.379)	0.395 (0.489)	0.43*** (0.06)
Mean error, conditional on misreporting	\$1.334 (1.850)	\$0.162 (0.828)	\$1.172*** (0.275)	\$1.01*** (0.303)	\$0.40 (0.379)	-\$0.10 (0.489)	\$0.50 (0.06)
N	68	146			271	91	

Notes. In columns (1), (2), (3), and (6), standard deviations are in parentheses; in columns (3), (4), and (7) robust standard errors are in parentheses and ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively. "Error" in the third row is computed as estimated toll - actual toll paid. In the MA Survey an estimate of the toll paid was eventually elicited from all but one of the respondents; however, in the NYNJ Survey, an estimate of the toll paid was only elicited for those who did not respond "don't know" or "refused." Thus, for the MA Survey, the sample in rows (2) and (3) includes all but one of the respondents in row (1), but for the NYNJ Survey, the sample in rows (2) and (3) includes only those respondents who did not report "don't know" in row (1). For the NYNJ survey, the cash toll was \$0.00 on peak and \$0.00 off peak. For the MA survey, the toll depended on the entrance and exit taken. The average toll paid was about \$1.15. Less than 10% of drivers in the MA survey sample drove on a portion of the Turnpike in which there are ETC discounts, and the results are not affected by omitting these drivers from the analysis. In column (4), covariates consist of age, age squared, median household income of ZIP code, dealer retail price for the driver's car (based on information from www.edmunds.com as of October 2007), and indicator variables for sex, whether the driver regularly pays a toll on a commute to work, and highest level of education (middle high school degree or less, college degree, or postgraduate degree, where "college degree" includes associate degrees, which were 10% of the college degree sample). Only published summary statistics (as opposed to the underlying microdata) are available for the NYNJ survey, so that the covariate-adjusted difference in means cannot be computed. In addition, the sample sizes for toll for the NYNJ survey had to be approximated based on information in the text on the total sample size (362) and the fraction of drivers that pay by ETC (74.8%). As a result, the standard errors for the NYNJ Survey are also approximated; approximated numbers are shown in italics. I calculated standard deviations for the binary response variables in the NYNJ Survey but there was not sufficient information available to calculate the standard deviation for the mean error (or the standard error of the difference in mean error).

(a) Table I: Survey evidence on driver awareness of tolls.



(b) Figure I: Distribution of ETC start dates.

6.2 Table II and Figure II: Which facilities adopt ETC and how ETC diffuses

Read:

- Table II shows that ETC adoption differs across regions and facility types. Adoption is highest in the Northeast and lowest in the West.
- Roads and bridges/tunnels adopt at similar rates, but roads that adopt tend to adopt earlier.
- Figure II shows gradual diffusion within facilities. ETC penetration rises quickly early on and approaches a steady-state level near 60 percent after roughly 14 years.

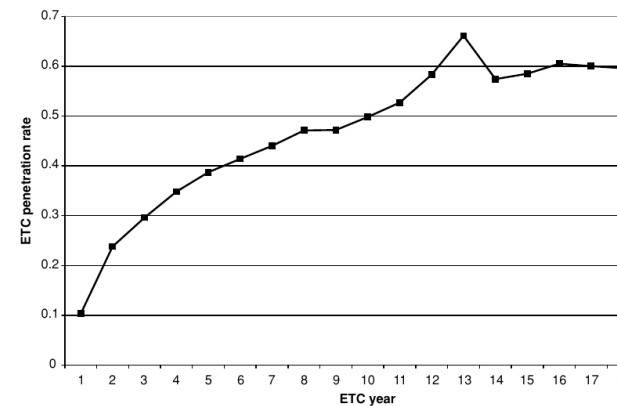
Economic message: ETC adoption is not random, so robustness checks are important. The diffusion pattern matters because the salience change strengthens gradually as more drivers pay electronically.

E-ZTAX: TAX SALIENCE AND TAX RATES

TABLE II
WHICH FACILITIES ADOPT ETC?

	Number of facilities	Probability of adopting ETC by 2005	Average adoptic date conditions on adoption
All	123	.71	1998.2
By facility type			
Roads	44	.70	1996.4
Bridges or tunnels	79	.71	1999.2

(a) Table II: Which facilities adopt ETC?



(b) Figure II: Within-facility ETC diffusion.

6.3 Tables III and IV: Demand elasticity and toll-rate effects

Read:

- Table III estimates how traffic responds to toll changes. Before ETC, driving is very inelastic with respect to tolls.
- The interaction between toll changes and ETC penetration is positive, meaning demand becomes less elastic in absolute value as ETC use rises.
- Table IV estimates the effect of ETC on toll rates. Once a facility has ETC, tolls grow faster.
- IV specifications using ETC penetration imply that a 10 percentage point increase in ETC penetration is associated with roughly a 5–6 percent toll increase.

Economic message: The demand result provides one mechanism: reduced salience lowers behavioral responsiveness. The toll-rate result shows the main policy effect: lower salience is associated with higher tolls.

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TABLE III
THE ELASTICITY OF TRAFFIC WITH RESPECT TO TOLLS

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log \text{ min. toll}_{it}$	-0.049 (0.015) [.004]	-0.058 (0.018) [.008]	-0.061 (0.019) [.009]	-0.057 (0.017) [.006]	-0.062 (0.039) [.145]	-0.060 (0.037) [.135]
$\Delta \log \text{ min. toll}_{it} * \text{ETC}_{it}$			0.134 (0.038) [.005]		0.141 (0.076) [.091]	
$\Delta \log \text{ min. toll}_{it} * \text{ETC}_{year_{it}}$				0.006 (0.001) [.002]		0.006 (0.003) [.062]
$\Delta \log \text{ min. toll}_{it} * \text{never_ETC}_i$			-0.071 (0.136) [.611]	-0.073 (0.131) [.588]	-0.009 (0.209) [.966]	-0.006 (0.205) [.976]
Mean dep. var.	0.049	0.042	0.043	0.042	0.040	0.039
# of states	21	12	12	12	12	12
# op. authorities	32	16	16	16	16	16
# of facilities	76	33	33	33	33	33

(a) Table III: Elasticity of traffic with respect to tolls.

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TABLE IV
IMPACT OF ETC ON TOLL RATES

	$\Delta \log$ min. toll (1)	$\Delta \log$ manual toll (2)	$\Delta \log$ toll (3)	$\Delta \log$ toll (4)	$\Delta \log$ min. toll (5)	$\Delta \log$ min. tol (6)
ETC_{it}	0.015 (0.006) [.018]	0.020 (0.006) [.004]	0.024 (0.012) [.061]			
ΔETC_{it}				0.623 (0.285) [.044]	0.557 (0.262) [.045]	0.501 (0.261) [.067]
ETCAdopt_{it}	-0.051 (0.035) [.158]	0.016 (0.032) [.622]	-0.033 (0.019) [.097]	-0.051 (0.035) [.166]	-0.105 (0.109) [.348]	-0.097 (0.108) [.380]
Mean dep. var.	0.020	0.022	0.017	0.017	0.020	0.020
# of states	24	24	17	17	24	24
# op. authorities	49	49	31	31	49	49
# facilities	123	123	70	70	123	123
<i>N</i>	5,079	5,079	2,875	2,751	4,815	4,815
Estimation	OLS	OLS	OLS	OLS	IV	IV
Sample restriction			No ETC discount	No ETC discount		

Notes. Table reports results of estimating (17) (columns (1)–(3)) and (19) (columns (4)–(6)). Column

(b) Table IV: Impact of ETC on toll rates.

6.4 Figure III: Time pattern of toll changes and ETC diffusion

Read:

- Figure III plots toll changes around ETC adoption together with ETC penetration.
- Toll rates are relatively flat before ETC adoption.
- Toll rates decline slightly in the first two years of ETC, consistent with initial ETC discounts.
- Afterward, tolls rise as ETC penetration rises.
- Panel B uses a more balanced sample of early adopters and shows a similar pattern.

Economic message: The timing supports the salience interpretation: tolls rise gradually as electronic payment becomes more common, not simply at the mechanical installation date.

E-ZTAX: TAX SALIENCE AND TAX RATES

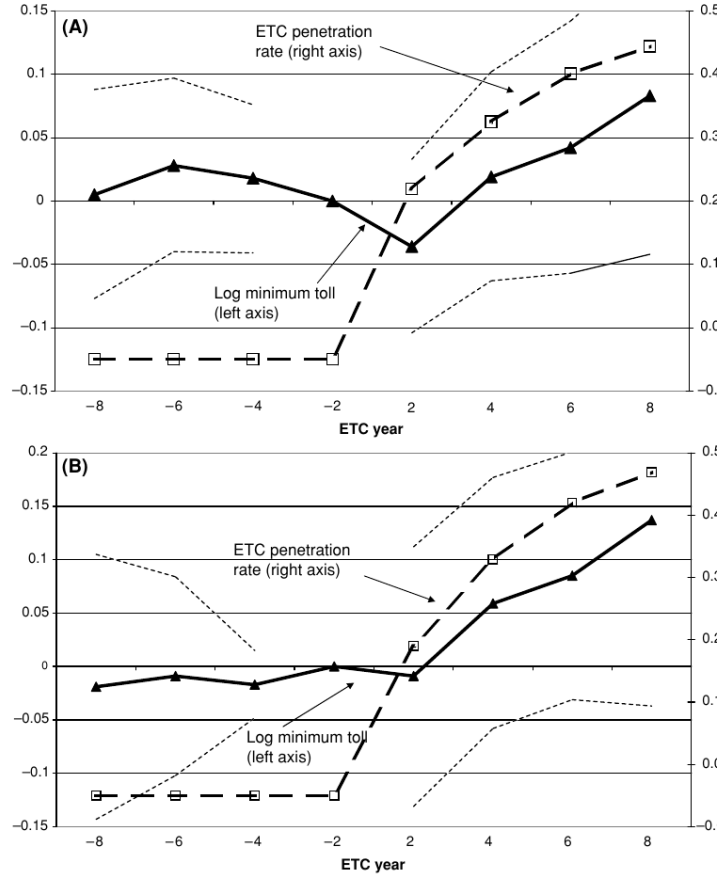


FIGURE III
Time Pattern of Toll Changes and ETC Diffusion

The solid black line shows the pattern of log minimum toll implied by estimates from (18a); the light dashed lines show the corresponding 95% confidence interval. The dark dashed line shows the pattern of the ETC penetration implied by estimating (18b). ETC year represents the number of years since before ETC adoption. The omitted category (ETC year -2 for (18a) and all years prior to ETC adoption for (18b)) is set to zero. Indicator variables for whether it is nine or more years after ETC adoption are included in the estimating equation but not graphed; in (4a) an indicator variable for whether it is nine or more years before ETC adoption is also included in the regression but not graphed. In Panel B the sample of ETC-adopting facilities is limited to those who adopted in 1990 or earlier. The upper end of the 95% confidence interval for the log minimum at eight years is not shown for scale reasons; it is 0.201 (full sample, A) and 0.201 (balanced panel, B). To enhance the readability of the graph, the 95% confidence interval on ETC penetration rate is not shown. For Panel A the upper and lower 95% confidence intervals for ETC penetration rate are as follows: (0.16, 0.267) for ETC year 2, (0.267, 0.484) for ETC year 4, (0.336, 0.565) for ETC year 6, (0.378, 0.610) for ETC year 8. For Panel B, the analogous confidence interval is (0.167, 0.333) for ETC year 2, (0.333, 0.485) for ETC year 4, (0.380, 0.550) for ETC year 6, (0.410, 0.615) for ETC year 8.

Figure III: Time pattern of toll changes and ETC diffusion.

6.5 Tables V and VI: Robustness and pretrends

Read:

- Table V shows that the toll increase after ETC is robust across region, facility type, facility age, facility fixed effects, and infrastructure controls.

- The estimated effect remains similar when comparing roads versus bridges/tunnels, old versus new facilities, and facilities with infrastructure histories.
- Table VI checks pretrends in traffic, revenue, and tolls before ETC adoption.
- There is no strong evidence of systematic changes in traffic, revenue, or tolls one to five years before adoption.

Economic message: These results address the concern that ETC adoption is merely correlated with preexisting toll-growth trends, congestion, revenue shocks, or infrastructure projects.

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TABLE VI
CHANGES IN TRAFFIC, REVENUE, AND TOLLS PRIOR TO ETC ADOPTION

	Dep. var.: $\Delta \log(\text{traffic})$		Dep. var.: $\Delta \log(\text{revenue})$		Dep. var.: $\Delta \log(\text{minimum toll})$	
	(1)	(2)	(3)	(4)	(5)	(6)
1–2 years before	–0.000		–0.009		0.004	
ETCAdopted _{it}	(0.007)		(0.016)		(0.013)	
	[.955]		[.599]		[.777]	
1–5 years before		0.013		0.006		0.009
ETCAdopted _{it}		(0.010)		(0.012)		(0.007)
		[.198]		[.601]		[.242]
ETCAdopt _{it}	–0.000	0.000	0.002	0.002	–0.051	–0.051
	(0.010)	(0.010)	(0.025)	(0.025)	(0.035)	(0.035)
	[.996]	[.978]	[.922]	[.930]	[.158]	[.162]
ETC _{it}	–0.006	–0.001	0.028	0.031	0.016	0.017
	(0.010)	(0.010)	(0.015)	(0.015)	(0.006)	(0.006)
	[.551]	[.959]	[.090]	[.058]	[.018]	[.008]
Mean dep. var	0.049		0.077		0.020	
# of states	21		13		24	

TABLE V
IMPACT OF ETC ON TOLL RATES: ROBUSTNESS ANALYSIS

	Baseline (1)	Northeast and midwest (2)	South and west (3)	Roads (4)	Bridges and tunnels (5)	Open after 1960 (6)	Open 1960 or before (7)	Facility fixed effects (8)	Facilities with infrastructure data	
									(9)	(10)
ETC _{it}	0.015 (0.006) [.018]	0.016 (0.010) [.141]	0.014 (0.005) [.030]	0.015 (0.008) [.067]	0.028 (0.010) [.015]	0.021 (0.010) [.065]	0.013 (0.007) [.079]	0.013 (0.007) [.072]	–0.055 (0.035) [.125]	–0.055 (0.035) [.124]
ETCAdopt _{it}	–0.051 (0.035) [.158]	–0.048 (0.054) [.399]	–0.044 (0.027) [.137]	–0.023 (0.063) [.719]	–0.086 (0.017) [.000]	0.051 (0.079) [.534]	–0.084 (0.025) [.005]	–0.053 (0.036) [.147]	0.014 (0.007) [.048]	0.014 (0.007) [.048]
INFRAADOP _{it}									0.017 (0.014) [.221]	
INFRA _{it}									–0.003 (0.007) [.659]	
Mean dep. var.	0.020	0.022	0.017	0.021	0.020	0.019	0.021	0.020	0.021	0.021
# of states	24	14	10	18	16	13	21	24	23	23
# op. authorities	49	28	21	24	31	20	39	49	46	46
# of facilities	123	68	55	44	79	43	77	123	115	115
N	5,079	3,008	2,071	1,692	3,387	1,380	3,609	5,079	4,712	4,712

Notes. Table reports results from estimating variants of (1) by OLS. The dependent variable is the change in the log minimum toll. All regressions include year fixed effects (not shown). Each opening authority receives equal weight. ETCAdopt_{it} is an indicator variable for whether facility *i* adopted ETC in year *t*. ETC_{it} is an indicator variable for whether facility has ETC; it is 1 in the year that ETC is adopted and in all subsequent years. Columns (2) and (3) limit the sample to, respectively, facilities in the northeast and midwest, and facilities in the south and west. Columns (4) and (5) limit the sample to, respectively, roads, and bridges or tunnels. Column (6) limit the sample to, respectively, facilities that opened after 1960 and facilities that opened in 1960 or earlier. Column (8) adds facility fixed effects to the right-hand side of (1). In columns (9) and (10) the sample is limited to the 115 facilities for which infrastructure data are available. INFRAADOP_{it} is an indicator variable for whether facility *i* started a new infrastructure project in year *t*. INFRA_{it} is an indicator variable for whether facility *i* has infrastructure data available.

(a) Table V: Robustness analysis of ETC effects on toll rates.

(b) Table VI: Changes in traffic, revenue, and tolls prior to ETC adoption.

6.6 Table VII: ETC and the politics of toll setting

Read:

- Table VII shows a political business cycle under manual toll collection: toll increases are smaller or less likely in election years.
- The interaction between ETC and election years is positive, especially for legislature-only election years.
- Under ETC, toll-setting is less sensitive to the election calendar.

Economic message: This is the key political-economy mechanism. ETC appears to reduce the political cost of raising tolls, consistent with the idea that less salient taxes weaken voter discipline.

TABLE VII
THE IMPACT OF ETC ON THE POLITICS OF TOLL SETTING

	$\Delta \log$ min toll (1)	Min toll raised? (2)	$\Delta \log$ min. toll (3)	Min toll raised? (4)	$\Delta \log$ min. toll (5)	Min toll raised? (6)
ETC _{it}	0.015 (0.006) [.018]	0.073 (0.024) [.006]	0.006 (0.009) [.507]	0.044 (0.022) [.042]	0.006 (0.009) [.494]	0.044 (0.022) [.042]
AnyElec Year _{st}			-0.016 (0.004) [.000]	-0.029 (0.010) [.003]		
GovElec Year _{st}					-0.016 (0.005) [.001]	-0.036 (0.012) [.002]
LegOnly ElecYear _{st}					-0.015 (0.005) [.005]	-0.021 (0.012) [.085]
AnyElec Year _{st} *ETC _{it}			0.017 (0.012) [.140]	0.055 (0.027) [.041]		
GovElec Year _{st} *ETC _{it}					0.004 (0.014) [.791]	0.016 (0.033) [.617]

Table VII: Impact of ETC on the politics of toll setting.

7 Short summary

- The paper studies whether tax salience affects equilibrium tax rates.
- ETC is used as a real-world salience shock because drivers paying electronically are less aware of the toll amount.
- Survey evidence confirms that ETC users know toll amounts much less accurately than cash users.
- A new panel of 123 U.S. toll facilities provides variation in ETC adoption dates and toll histories.
- Traffic becomes less elastic with respect to tolls as ETC penetration rises.
- Tolls rise faster after ETC adoption.
- Once ETC reaches steady-state penetration, tolls are estimated to be 20–40 percent higher than they would have been under manual collection.
- The time pattern of toll increases tracks ETC diffusion.
- Robustness checks find no strong pretrends or infrastructure-project explanation.
- The election-year suppression of toll increases disappears under ETC, suggesting lower political costs of less visible taxes.

8 Conclusion

8.1 Core conclusion

Finkelstein shows that reducing the salience of a tax instrument can increase equilibrium tax rates. In the toll context, electronic payment makes tolls less visible to drivers, and facilities that adopt ETC subsequently raise tolls more than they otherwise would have.

8.2 Economic intuition

When drivers pay cash, the toll is salient: they count money, hand it over, and directly observe the price. When they pay electronically, the transaction is automatic and attention to the toll amount falls. This reduced salience weakens the behavioral response to toll increases and lowers the political cost of raising tolls. Both channels make higher tolls more likely.

8.3 Incremental contribution revisited

The paper’s main contribution is to move the tax-salience literature from individual misperception to political equilibrium. It shows that the design of the tax instrument can affect not only consumer behavior but also the tax rate set by governments.

8.4 Implications

The findings matter for debates about withholding, VATs, electronic payments, and automatic collection systems. Less visible taxes may reduce compliance costs and administrative burdens, but they may also reduce voter monitoring and permit higher tax rates.

8.5 Limitations

The setting is tolls, which are small relative to income and administered by toll authorities, so the effect need not generalize mechanically to large taxes such as income taxes. The design is observational and relies on the timing of ETC adoption not being correlated with unobserved toll-setting shocks. The paper addresses this with extensive robustness checks, but no natural experiment is perfect.

8.6 Takeaway

The enduring takeaway is that tax salience is politically consequential. A less visible tax system can produce higher taxes because citizens respond less strongly both as consumers and as voters.